

STAT210 2025: Midterm Exam (60 points)

October 30, 2025

Problem 1 (15 points). Consider the multiple linear regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where $\mathbf{y}$ is an $n \times 1$ vector of responses,

$\mathbf{X}$ is an $n \times (p + 1)$ design matrix with full column rank (the first column is a vector of 1's),

$\boldsymbol{\beta}$ is a $(p + 1) \times 1$ vector of regression coefficients,

and $\boldsymbol{\varepsilon}$ is an $n \times 1$ vector of random errors satisfying

$$E(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n.$$

The least squares estimator of $\boldsymbol{\beta}$ is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y},$$

and the fitted values and residuals are given by

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}, \quad \mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}.$$

Let

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

denote the hat (or projection) matrix.

(a) (5 points)

Show that $\mathbf{H}$ is a projection matrix; that is, verify that $\mathbf{H}$ is symmetric and idempotent:

$$\mathbf{H}' = \mathbf{H}, \quad \mathbf{H}^2 = \mathbf{H}.$$

(b) (5 points)

Show that $\mathbf{I}_n - \mathbf{H}$ is also a projection matrix and explain in words what it projects onto.

(c) (5 points)

Let $\hat{\boldsymbol{\beta}}$ and $\mathbf{e}$ be defined as above.

We know that

$$\hat{\boldsymbol{\beta}} = \mathbf{H}\mathbf{y}, \quad \mathbf{e} = \mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I}_n - \mathbf{H})\mathbf{y}.$$

Use the above result to find the covariance between $\hat{\boldsymbol{\beta}}$ and $\mathbf{e}$, and then show that they are uncorrelated.

Hint: Use $\text{Cov}(\mathbf{A}\mathbf{y}, \mathbf{B}\mathbf{y}) = \mathbf{A} \text{Var}(\mathbf{y}) \mathbf{B}'$ and $\text{Var}(\mathbf{y}) = \sigma^2 \mathbf{I}_n$.

Problem 2 (20 points). For the `mtcars` dataset, we first regress `mpg` on `wt`, `hp`, and `disp`. The coefficient table is provided below.

```
#mtcars: mpg vs disp, wt, hp
fit1 <- lm(mpg ~ wt + hp + disp, data = mtcars)

# model fit1:
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  37.105505   2.110815  17.579  < 2e-16 ***
wt           -3.800891   1.066191  -3.565  0.00133 **
hp           -0.031157   0.011436  -2.724  0.01097 *
disp         -0.000937   0.010350  -0.091  0.92851
```

(a) (5 points)

Based on this model (`fit1`), interpret the coefficient of `wt` in the context of the problem. Provide a 95% confidence interval for this coefficient (here $n = 32$; you may leave the interval as a function of $t_{df,0.975}$ with the appropriate degrees of freedom).

(b) (5 points)

The unit of `wt` is 1000 lbs. Suppose we want to change the unit of `wt` to kilograms ($1 \text{ kg} = 2.20462 \text{ lbs}$). Provide the new coefficient and a 95% confidence interval for `wt` in the new model with `wt` in kilograms. You can leave your answer unsimplified but be sure to explain how you obtained these values.

(c) (5 points)

Consider the model with the `disp` as the only predictor:

```
fit2 <- lm(mpg ~ disp, data = mtcars)
```

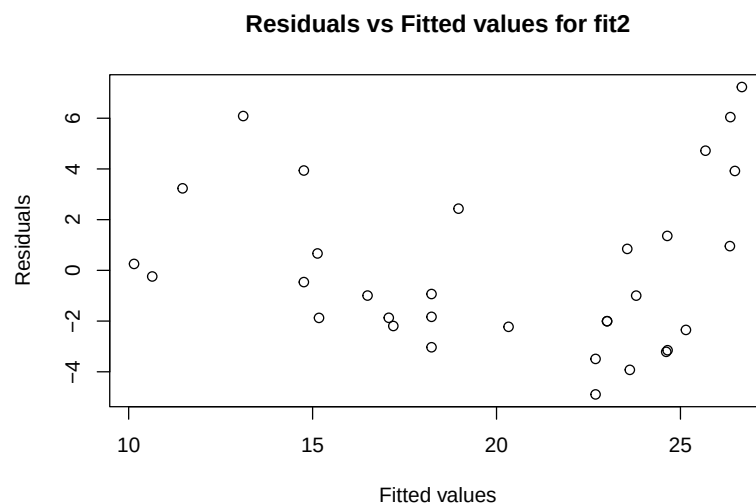
Its coefficient table is

```
# model fit2:
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  29.599855   1.229720  24.070  < 2e-16 ***
disp         -0.041215   0.004712  -8.747  9.38e-10 ***
```

Compared the results for `disp` in `fit1` and `fit2`. Does the difference in p-values indicate a contradiction? Explain why or why not.

(d) (5 points)

Here is the residual vs fitted value plot for `fit2`:



Based on this plot, explain whether the linear regression model is appropriate. If not, suggest how it might be improved (e.g., transformation, addressing nonlinearity or heteroscedasticity).

Problem 3 (25 points). Consider the multiple linear regression model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon,$$

where y is the unemployment rate and x_1 : percentage of population with a bachelor's degree, x_2 : per capita income, and x_3 : percentage of population over 65 years old.

Based on the following ANOVA table, answer each of the questions

```
## Analysis of Variance Table
##
## Response: unemployment (y)
##           Df Sum Sq Mean Sq  F value    Pr(>F)
## bachelor (x1)  1   702.05   702.05    182.0089 < 2e-16 ***
## income (x2)    1    13.54    13.54     3.5099  0.06167 .
## X65plus (x3)   1     2.17     2.17     0.5631  0.45343
## Residuals     436 1681.76    3.86
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(a) (5 points)

What is the sample size n ? what is the value of $\sum_{i=1}^n (y_i - \bar{y})^2$?

(b) (5 points)

In the ANOVA table, the p-value for X65plus is 0.45343. State the null and alternative hypotheses being tested, and explain what this p-value means in the context of the problem.

(c) (5 points)

Find an F statistic whose numerator degrees of freedom is 3. State the hypotheses being tested, its distribution under H_0 , and calculate the value of the F-statistic.

(d) (5 points)

You decide to fit a smaller model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon.$$

Based on the ANOVA table provided above, construct the ANOVA table of the reduced model.

(e) (5 points)

The coefficient of determination R^2 between each predictor and unemployment quantifies their marginal strength of correlation. Based on the ANOVA model, which of the following three R^2 can be computed? Find the value(s) and show your work.

- R^2 between unemployment and bachelor
- R^2 between unemployment and income
- R^2 between unemployment and X65plus